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GENERAL INTRODUCTION

As global population and consumer demand continue to grow, addressing the environmental impact of textile production has become increasingly important. The textile industry has a considerable environmental footprint. Additionally, textile production requires significant amounts of raw materials and land resources. In 2022, European household textile consumption alone used approximately 144,000 km^2 of land (about 323 m^2 per person) [1]. In terms of raw materials, around 175 million tonnes of primary materials (including fossil fuels, cotton fibers, and chemicals) were used for textile products for EU households in 2020, equivalent to roughly 391 kg per person [2]. The industry generates millions of tonnes of textile waste annually, with production having doubled in recent years while garment usage duration has decreased significantly. Recycling rates in the textile industry remain low, with only a small percentage of resources coming from recycled materials. This highlights the need for sustainable practices and circular economy approaches in textile manufacturing. Life Cycle Assessment (LCA) has become an important tool for evaluating and improving the environmental performance of textile products. LCA provides a framework for quantifying environmental impacts throughout a product's entire life cycle, from raw material extraction to end-of-life disposal. This study presents a Life Cycle Assessment of yarn production processes, focusing on analyzing the environmental impacts of different production methods and fiber compositions. This study identifies environmental hotspots in yarn manufacturing by assessing each stage of the spinning process for both virgin and recycled fibers. It compares the environmental performance of different spinning technologies, including seasonal variations, and evaluates the impacts of using recycled versus conventional materials. The results provide guidance for improving the sustainability of yarn production and supporting more environmentally responsible textile manufacturing.

Chapter3 : Yarn Manufacturing and Waste Management

1 Introduction

This chapter presents an overview of the yarn production processes and waste management practices at SITEX Sousse, focusing specifically on the operations considered in this study. The analysis is limited to carded yarn production, while combed yarn processes are excluded. The chapter describes the methods used for producing virgin and recycled cotton yarn, including ring spinning, open-end spinning. It also presents the main categories of production waste, along with the supporting systems such as air treatment, electrical transformers, and compressors, which are essential for establishing the context of the LCA comparison.

2 Overview of Yarn Production Processes

This section summarizes the main yarn production methods used in this study, including virgin and recycled processes.

2.1 Virgin Yarn Production Processes

2.1.1 Conventional Spinning (Ring Yarn)

The production of conventional ring-spun yarn begins with the opening of virgin cotton fiber bales using the Blendomat machine. This is followed by the opening stage, where the fibers are loosened, cleaned and prepared for further processing. At SITEX Sousse, the opening stage consists of three parallel routes: A, B, and C as shown in Appendix A.3. For conventional yarn production, Route C is most commonly used as it is equipped with a Vision Shield system capable of detecting fiber color through image processing. This system allows for the removal of yellowish cotton fibers, improving the visual quality of the final product. Route C includes a specific sequence of machines, including ventilation systems, multi-chamber mixers, condensers, CVT4 cleaners and dedusting units. After the opening stage, the fibers are fed into carding machines to individualize and straighten them into a uniform sliver. This sliver then undergoes two drawing operations to enhance fiber alignment and reduce thickness variations. The drawn sliver is subsequently processed in a roving frame, where a slight twist is applied to produce an intermediate strand. Finally, the roving enters the ring spinning machines, where the yarn is spun to the desired count and twist. The finished yarn is then wound onto cones in preparation for downstream use. The full production flow is presented in Appendix A.5.

2.1.2 Open-End (OE) Spinning

The initial stages of open-end yarn production blending, opening and carding are identical to those used in conventional ring spinning. However, after the drawing operations, the process diverges. Instead of proceeding to the roving and ring spinning stages, the drawn sliver is directly fed into Autocoro open-end spinning machines, where yarn is produced through rotor spinning technology. This method eliminates the need for intermediate roving, enabling faster production. The complete flow is included in Appendix A.5.

2.2 Recycled Yarn Production Processes

2.2.1 Shredded Yarn Process

The shredded yarn process begins with the intake of textile waste materials such as fabric remnants generated during the fabric cutting stage, selvedge edges from weaving machines or dyed yarns from the dyeing stage, sourced from various suppliers, including SITEX Ksar Hellal. The different types of input materials are shown in Figure 3.1.

Figure 3.1: Some of Different Input Materials Used in the EXEL Machine

These materials are first directed to the shredding room, where they undergo a series of operations: cutting, lubrication and temporary storage in blending chambers, as illustrated in Figure 3.2. Next, the pre-processed material is fed into the shredding machine, known as Exel, as shown in Figure 3.2, which uses a series of cylinders with increasing speeds and garniture to mechanically break down the fibers. The Exel unit is connected to a compressor and a bale press, which compacts the shredded fibers into bales. These bales are then transferred to a pre-drawing machine, functioning as a bale opener and a modified carding unit, also shown in Figure 3.2. The resulting fibers are either stored or directly integrated into the spinning process. They are typically blended with virgin cotton or polyester and processed through either the conventional ring spinning or the open-end spinning line. The complete process flow is included in Appendix A.5.

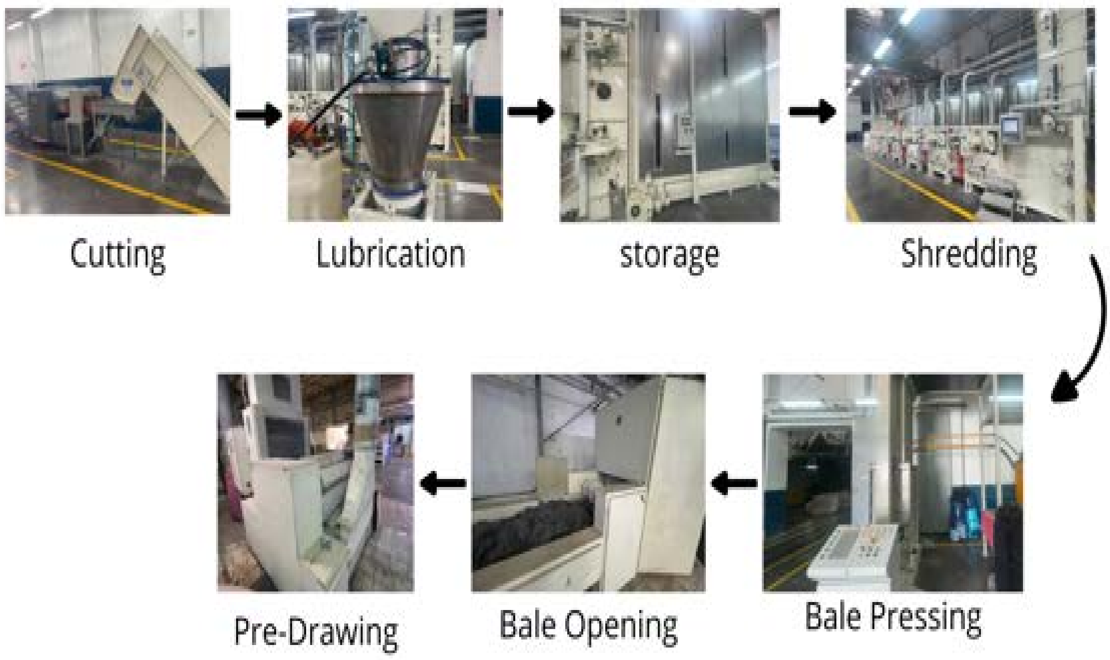

Figure 3.2: Shredding Process

2.2.2 RL Yarn Process

Recycled RL yarn is produced from waste generated during the carding stage, primarily composed of dead fibers extracted by the internal aspiration system of the carding machines, as shown in Figure 3.3. This RL material is considered a co-product of the process.

Figure 3.3: RL waste from recycled cotton

These fibers are collected via pneumatic suction and transported through ducts to a centralized compression unit, where they are compacted into bales, as shown in Figure 3.4.

Figure 3.4: RL Waste Compression Unit

The RL bales are then directed to the RL processing line, which consists of a series of machines designed to reprocess this material. The line typically begins with a bale opener, followed by a modified carding unit equipped with finer cylinders for delicate fiber treatment. The material then passes through a CVT4 cleaning system to eliminate remaining impurities before being recompressed into bales, as presented in Figure 3.5. These recycled bales can either be stored or directly integrated into the spinning process. They are commonly blended with virgin cotton or polyester and processed through either the conventional ring spinning or open-end spinning line, depending on the desired yarn characteristics. As a co-product of the carding stage, RL yarn represents a way to recover material that would otherwise be considered waste. The complete process is illustrated in Appendix A.5.

Figure 3.5: RL Line Unit

2.2.3 F3 Yarn Production

F3 yarn is produced from internal production waste collected across various stages of the spinning process, as shown in Appendix A.5. This F3 material is considered a co-product of the spinning process. These materials are gathered in the F3 unit, sorted by type of waste, and then passed through a CVT4 cleaning machine, as shown in Figure 3.6. After cleaning, the fibers are compacted using a bale press, illustrated in Figure 3.7. The resulting bales can either be stored or reintegrated into the spinning process after being blended with other fibers.

Figure 3.6: CVT4 Cleaning Machine In The F3 Unit

Figure 3.7: Bale Press In The F3 Unit

3 Characterization of production Waste

Production waste in the spinning process can be broadly classified into two categories: recoverable waste, which can be reintegrated into the production cycle and non-recoverable waste.

3.1 Recoverable Waste

3.1.1 RL fibers

This waste consists primarily of dead fibers extracted during the carding process, as shown in Figure 3.3. The method of collection is explained in detail in Section 2.2.2 . RL waste is mainly composed of short, damaged fibers and neps. These byproducts result from intensive mechanical processing and are generally unsuitable for fine count yarns due to their low spinnability and weak fiber cohesion. However, when blended with stronger fibers, they serve as a sustainable raw material for producing coarse yarns.

3.1.2 Production Waste

Production waste encompasses recoverable fiber residues generated throughout various stages of the spinning process, excluding RL fibers. As explained in Section 2.2.3, this waste is collected at points such as carding, drawing, roving, spinning, winding and quality control. Operators place the waste into designated containers behind each machine, as shown in Figure 3.8, which are then transferred to the F3 unit for sorting, cleaning and reuse in yarn manufacturing.

Figure 3.8: Waste Container

The types of waste collected include:

- Waste from the carding stage, generated when rubans are cut during processing.
- Waste from drawing, also resulting from ruban cuts.
- Roving waste extracted by the machine's aspiration system and sliver cuts.
- Waste known as Captafibre, which consists of sliver cuts generated during spinning and winding, is manually collected by operators. Additional aspiration waste is extracted through suction systems, as shown in Figure 3.9(a) and (b).
- Remaining slivers left after cone completion are retrieved by a specialized machine that rotates the cone in the opposite direction of the original sliver winding. This mechanism, shown in Figure 3.9 (c), unwinds and releases the residual fibers efficiently into large containers for collection, as illustrated in Figure 3.9 (d).
- Residual waste trapped on the drafting cylinders in spinning machines, as shown in Figure 3.9 (e), is periodically eliminated using a dedicated cleaning machine shown in Figure 3.9 (f).
- Measurement waste generated during quality checks in the control room, such as linear density control, is depicted in Figure 3.9 (g).

6.2 Cross-check with Literature Results

To contextualize the results of this study, a comparison is made with the findings of Liu et al. (2020) [**liu2020**] [12], who performed a detailed life cycle assessment (LCA) on virgin and recycled cotton yarns. Their study, titled *"Could the recycled yarns substitute for the virgin cotton yarns: a comparative LCA"*, was published in *The International Journal of Life Cycle Assessment*. The goal of Liu et al.'s study was to evaluate whether recycled cotton yarns could serve as a sustainable alternative to virgin cotton yarns by comparing their environmental impacts across multiple categories. The system boundary adopted in their LCA is cradle-to-gate. The functional unit was defined as 1 tonne of spun yarn, whereas in the present study, the functional unit is 1 kilogram. Therefore, Liu et al.'s results were divided by 1,000 to enable a consistent comparison on a per-kilogram basis. Liu et al. assessed a broad set of environmental indicators based on the ReCiPe method, including global warming potential (GWP), water depletion, fossil depletion, land occupation, human toxicity, ecotoxicity, and other mid-point indicators. Their results showed that recycled cotton yarn consistently outperformed virgin cotton yarn across nearly all categories, particularly in climate change, water depletion, and land use. To benchmark the environmental performance of the yarns assessed in this work, Table 4.32 compares the key results from Liu et al. with four yarn types from the current study: M2 Conv, M3 Conv, M6 Conv, and 100% RC Conv.

Table 4.32: Comparison of environmental impacts with literature results

Yarn / Study	GWP (kg CO_2-eq/kg)	Water Use (m^3/kg)	Fossil Scarcity (kg oil-eq/kg)	Land Use ($m^2\cdot$yr/kg)	Spinning Method	Composition	Source
M2 Conv (This study)	3.73	0.696	0.854	3.62	Ring	50% cotton / 50% recycled cotton	—
M3 Conv (This study)	5.04	1.52	0.766	7.95	Ring	100% virgin cotton	—
M6 Conv (This study)	8.71	0.423	3.15	2.13	Ring	45% shredded textile / 25% virgin cotton / 30% PES	—
100% RC Conv (This study)	3.01	0.01	1.04	0.00	Ring	100% recycled cotton	—
Liu (2020) – Recycled	4.38	0.583	1.10	0.00	Ring	100% recycled cotton	Liu et al. (2020)
Liu (2020) – Virgin	11.00	3.514	1.24	5.80	Ring	100% virgin cotton	Liu et al. (2020)

As presented in Table 4.32, the environmental impacts of M3 Conv (100% virgin cotton) are lower than those reported by Liu et al. (2020) for virgin cotton yarn in Global Warming Potential (5.04 vs. 11.00 kg CO_2-eq/kg) and Fossil Resource Scarcity (0.766 vs. 1.24 kg oil-eq/kg). This reduction is attributed to the use of allocated virgin cotton in the present study, which accounts for the recycled fraction in the allocation and thus lowers the environmental burdens compared to studies considering only virgin cotton. However, M3 Conv shows higher Land Use (7.95 vs. 5.80 $m^2\cdot$yr/kg) and slightly lower Water Use

(1.52 vs. 3.514 m^3/kg), likely due to regional differences in cultivation and background LCA datasets. The 100% recycled cotton yarn from this study (100% RC Conv) exhibits the lowest Water Use (0.01 m^3/kg) among all yarns, a low GWP (3.01 kg CO_2-eq/kg), and negligible Land Use (0.00 $m^2 \cdot yr$/kg). Its Fossil Resource Scarcity (1.04 kg oil-eq/kg) is slightly higher than M2 Conv (0.854 kg oil-eq/kg) but remains substantially lower than M6 Conv (3.15 kg oil-eq/kg). This confirms that fully recycled cotton yarn provides significant environmental benefits, especially in water consumption and land occupation, compared with both virgin cotton yarns and partially recycled blends.

Overall, the comparison reinforces that increasing recycled cotton content generally reduces environmental impacts, while highlighting that fiber blends containing polyester or other materials may create trade-offs in certain categories, such as Fossil Resource Scarcity and Global Warming Potential. These results underscore the importance of balanced material choices in sustainable yarn production.

7 Conclusion

In conclusion, the comparison with literature highlights the environmental advantages of incorporating recycled materials in yarn production. The findings reinforce the potential of alternative blends, such as M2 Conv and M6 Conv, to reduce environmental burdens in specific impact categories. This underscores the importance of thoughtful material selection and supports further development of circular strategies in the textile industry.

GENERAL CONCLUSION

This Life Cycle Assessment of yarn production processes at S.I.T.E.X. successfully quantified environmental impacts and identified key sustainability hotspots in textile manufacturing. The study compared Open-End (OE) and conventional spinning methods across multiple environmental impact categories. The research revealed that production method selection, fiber composition, and seasonal variations significantly influence yarn production's environmental footprint. Energy consumption, waste generation, and water usage vary considerably between different production configurations. The integration of recycled fibers showed promising potential for reducing environmental burden, though quality requirements of recycled materials affect both environmental benefits and production efficiency. Seasonal comparisons highlighted the critical importance of optimizing heating, ventilation, and air conditioning systems, which significantly impact environmental performance throughout the year. Process optimization should focus on improving energy efficiency in heating and cooling systems. Waste minimization through advanced waste recovery and recycling systems is essential. Technology integration requires evaluating emerging spinning technologies. Supply chain assessment should extend LCA boundaries to include upstream and downstream processes. Continuous monitoring systems need to be established to track regular environmental performance. This study addresses the textile industry's environmental challenges, where an additional $20-30 billion annually is needed for sustainable transformation. By identifying specific improvement areas, this research supports industry efforts to reduce environmental impact while maintaining production efficiency. The findings provide a foundation for evidence-based decision-making in sustainable textile manufacturing and contribute to the industry's transition toward environmentally responsible production practices that meet growing global demand while preserving resources for future generations.

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